

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 2: Input & Output Characteristics of BJT in
Common Emitter (CE), Common Base (CB), and Common
Collector (CC) Configurations

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1. Aim

1. To study and plot the input and output characteristics of a BJT (BC547 NPN Transistor) in the **Common Emitter (CE)** configuration, and to determine the common-emitter current gain (β), input resistance (r_i), and output resistance (r_o).
2. To study and plot the input and output characteristics of a BJT in the **Common Base (CB)** configuration, and to determine the common-base current gain (α), input resistance (r_i), and output resistance (r_o).
3. To study and plot the input and output characteristics of a BJT in the **Common Collector (CC)** configuration, and to determine the common-collector current gain (γ), input resistance (r_i), and output resistance (r_o).
4. To verify the mathematical relationships between current gains: $\beta = \frac{\alpha}{1-\alpha}$ and $\gamma = \beta + 1$ using measured parameters.

2. Apparatus and Software Tools Required

Table 1: Equipment, Components, and Software Specifications

S.No.	Component / Equipment	Specification	Quantity
1	NPN Transistor	BC547 (TO-92 package)	1
2	Resistors	1 k Ω , 10 k Ω	1 each
3	Regulated DC Power Supplies	Variable 0–30 V Dual Output	2
4	Digital Ammeters / Microammeters	0–500 μ A, 0–10 mA	As required
5	Digital Voltmeters	0–1 V, 0–30 V	As required
6	Simulation Software	LTspice XVII	–

3. Overall Theoretical Background

A Bipolar Junction Transistor (BJT) is a three-terminal, current-controlled semiconductor device consisting of two PN junctions (emitter-base junction J_{EB} and collector-base junction J_{CB}). The behavior of the transistor in any circuit configuration depends primarily on the biasing conditions of these two junctions:

- **Cut-off Region:** Both J_{EB} and J_{CB} are reverse-biased. Transistor is OFF, $I_C \approx 0$.

- **Active Region:** J_{EB} is forward-biased and J_{CB} is reverse-biased. Transistor operates as an amplifier.
- **Saturation Region:** Both J_{EB} and J_{CB} are forward-biased. Transistor acts as a closed switch, $V_{CE} \approx V_{CE(\text{sat})}$.

Depending on which terminal is kept common to both input and output ports, BJTs are analyzed in three distinct configurations: Common Emitter (CE), Common Base (CB), and Common Collector (CC).

4. Part A & B: Common Emitter (CE) Configuration

4.1 Theoretical Background

In the Common Emitter (CE) configuration, the emitter terminal is common to both the input (base-emitter) and output (collector-emitter) circuits.

- **Input Characteristics:** Plot of Base Current (I_B) vs Base-Emitter Voltage (V_{BE}) at constant Collector-Emitter Voltage (V_{CE}). It behaves like a forward-biased diode curve. As V_{CE} increases, the effective base width decreases (Early effect / Base-width modulation), shifting the input curve slightly to the right.
- **Output Characteristics:** Plot of Collector Current (I_C) vs V_{CE} at constant I_B . In the active region, $I_C = \beta I_B$, showing flat curves with high output resistance.
- **Key Parameters:**

$$\text{Input Resistance: } r_i = \left. \frac{\Delta V_{BE}}{\Delta I_B} \right|_{V_{CE}=\text{const}} \quad (\approx 1 \text{ k}\Omega - 5 \text{ k}\Omega) \quad (1)$$

$$\text{Output Resistance: } r_o = \left. \frac{\Delta V_{CE}}{\Delta I_C} \right|_{I_B=\text{const}} \quad (\approx 50 \text{ k}\Omega - 100 \text{ k}\Omega) \quad (2)$$

$$\text{Current Gain: } \beta = \left. \frac{\Delta I_C}{\Delta I_B} \right|_{V_{CE}=\text{const}} \quad (\approx 50 - 300) \quad (3)$$

4.2 Circuit Schematic and Simulation Waveforms

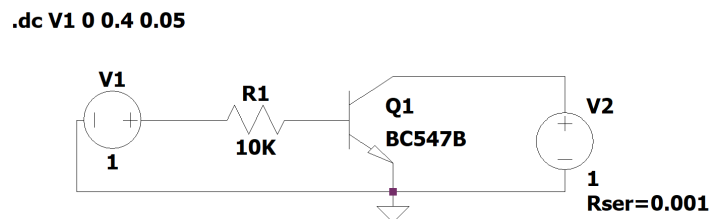
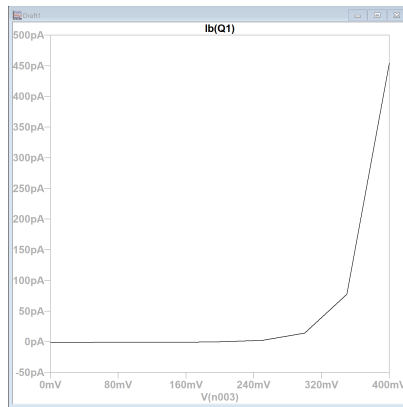
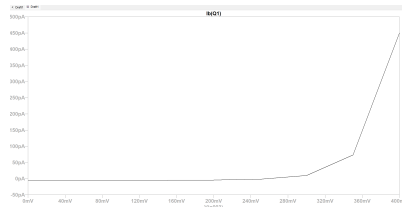
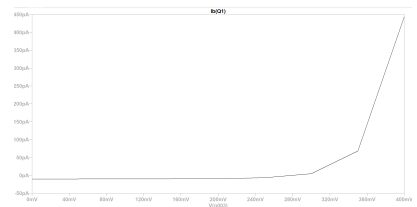
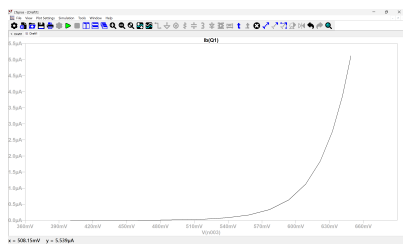
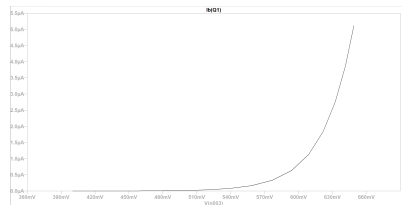
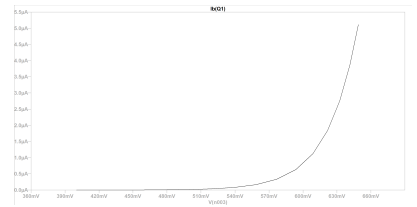
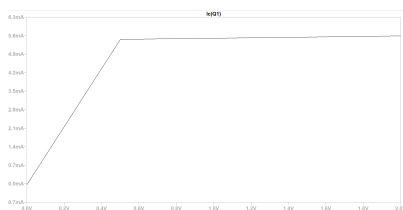
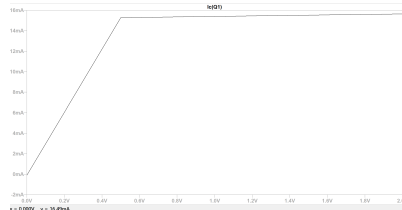
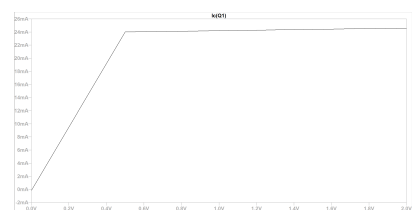
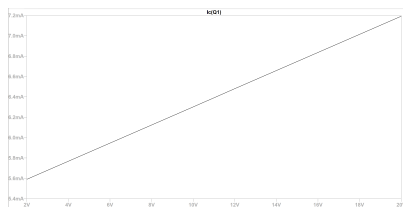
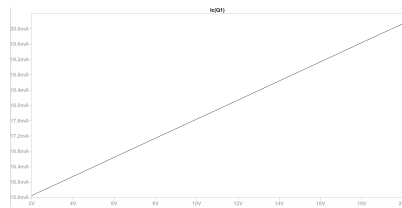
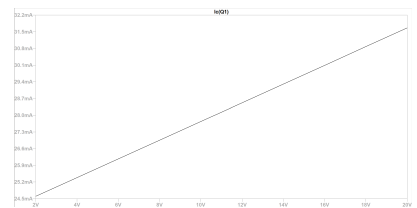


Figure 1: LTspice Circuit Schematic for Common Emitter (CE) Configuration.

4.2.1 CE Input Characteristics Simulation

(a) 0–0.4 V ($V_{CE} = 1$ V)(b) 0–0.4 V ($V_{CE} = 5$ V)(c) 0–0.4 V ($V_{CE} = 10$ V)(d) 0.4–0.7 V ($V_{CE} = 1$ V)(e) 0.4–0.7 V ($V_{CE} = 5$ V)(f) 0.4–0.7 V ($V_{CE} = 10$ V)Figure 2: LTspice Simulation of CE Input Characteristics (I_B vs V_{BE}) for $V_{CE} = 1$ V, 5 V, 10 V.

4.2.2 CE Output Characteristics Simulation

(a) $V_{CE} = 0$ –2V, $I_B = 20\mu\text{A}$ (b) $V_{CE} = 0$ –2V, $I_B = 60\mu\text{A}$ (c) $V_{CE} = 0$ –2V, $I_B = 100\mu\text{A}$ (d) $V_{CE} = 2$ –20V, $I_B = 20\mu\text{A}$ (e) $V_{CE} = 2$ –20V, $I_B = 60\mu\text{A}$ (f) $V_{CE} = 2$ –20V, $I_B = 100\mu\text{A}$ Figure 3: LTspice Simulation of CE Output Characteristics (I_C vs V_{CE}) in Saturation and Active Regions.

4.3 Experimental Observation Tables (Hardware Data)

Table 2: CE Input Characteristics Observation Table (I_B vs V_{BE})

S.No.	V_{BE} (V)	I_B (μA) at $V_{CE} = 1\text{V}$	I_B (μA) at $V_{CE} = 5\text{V}$	I_B (μA) at $V_{CE} = 10\text{V}$
1	0.00			
2	0.10			
3	0.20			
4	0.30			
5	0.40			
6	0.50			
7	0.55			
8	0.60			
9	0.62			
10	0.64			
11	0.66			
12	0.68			
13	0.70			

Table 3: CE Output Characteristics Observation Table (I_C vs V_{CE})

S.No.	V_{CE} (V)	I_C (mA) at $I_B = 20 \mu\text{A}$	I_C (mA) at $I_B = 60 \mu\text{A}$	I_C (mA) at $I_B = 100 \mu\text{A}$
1	0.0			
2	0.5			
3	1.0			
4	1.5			
5	2.0			
6	4.0			
7	6.0			
8	8.0			
9	10.0			
10	15.0			
11	20.0			

5. Part C & D: Common Base (CB) Configuration

5.1 Theoretical Background

In the Common Base (CB) configuration, the base terminal is grounded and common to both input (emitter-base) and output (collector-base) circuits.

- **Input Characteristics:** Plot of Emitter Current (I_E) vs Emitter-Base Voltage (V_{EB}) at constant V_{CB} . J_{EB} is forward-biased, yielding a very sharp exponential diode curve. Increasing V_{CB} accelerates minority carrier extraction across the base, slightly increasing I_E for a given V_{EB} .
- **Output Characteristics:** Plot of Collector Current (I_C) vs V_{CB} at constant I_E . Once $V_{CB} > 0$ V, $I_C \approx \alpha I_E$. The curves are almost perfectly flat, indicating extremely high output resistance.
- **Key Parameters:**

$$\text{Input Resistance: } r_i = \left. \frac{\Delta V_{EB}}{\Delta I_E} \right|_{V_{CB}=\text{const}} \quad (\text{Very Low: } \approx 10\text{--}50 \, \Omega) \quad (4)$$

$$\text{Output Resistance: } r_o = \left. \frac{\Delta V_{CB}}{\Delta I_C} \right|_{I_E=\text{const}} \quad (\text{Very High: } \approx 1 \, \text{M}\Omega) \quad (5)$$

$$\text{Current Gain: } \alpha = \left. \frac{\Delta I_C}{\Delta I_E} \right|_{V_{CB}=\text{const}} \quad (< 1, \text{ typically } 0.95\text{--}0.99) \quad (6)$$

5.2 Circuit Schematic and Simulation Waveforms

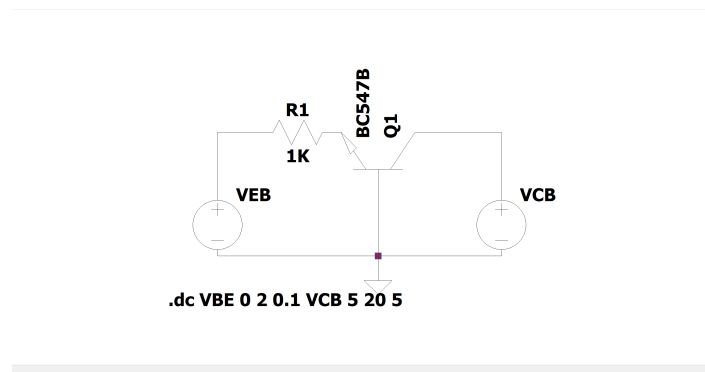
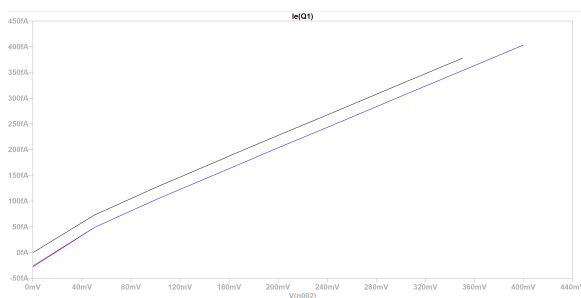
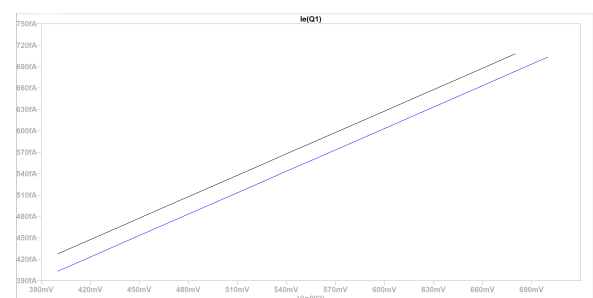


Figure 4: LTspice Circuit Schematic for Common Base (CB) Configuration.

5.2.1 CB Input Characteristics Simulation



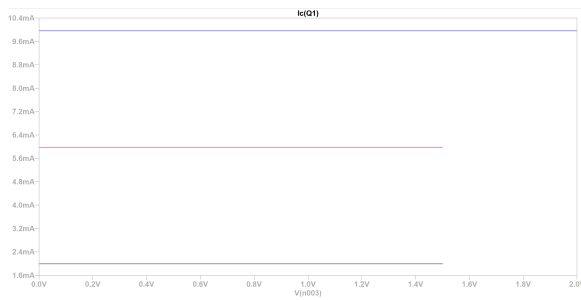
(a) Low Voltage Range ($V_{EB} = 0\text{--}0.4$ V)



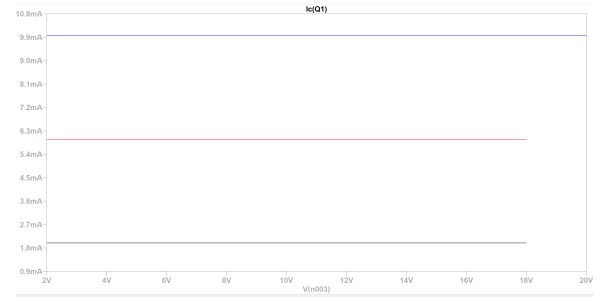
(b) Conduction Range ($V_{EB} = 0.4\text{--}0.7$ V)

Figure 5: LTspice Simulation of CB Input Characteristics (I_E vs V_{EB}) for various V_{CB} values.

5.2.2 CB Output Characteristics Simulation



(a) Knee / Transition Region ($V_{CB} = 0\text{--}2\text{ V}$)



(b) Active Region ($V_{CB} = 2\text{--}20\text{ V}$)

Figure 6: LTspice Simulation of CB Output Characteristics (I_C vs V_{CB}) for $I_E = 2\text{ mA}, 6\text{ mA}, 10\text{ mA}$.

5.3 Experimental Observation Tables (Hardware Data)

Table 4: CB Input Characteristics Observation Table (I_E vs V_{EB})

S.No.	V_{EB} (V)	I_E (mA) at $V_{CB} = 0\text{V}$	I_E (mA) at $V_{CB} = 5\text{V}$	I_E (mA) at $V_{CB} = 10\text{V}$
1	0.00			
2	0.10			
3	0.20			
4	0.30			
5	0.40			
6	0.50			
7	0.55			
8	0.60			
9	0.62			
10	0.64			
11	0.66			
12	0.68			
13	0.70			

Table 5: CB Output Characteristics Observation Table (I_C vs V_{CB})

S.No.	V_{CB} (V)	I_C (mA) at $I_E = 2\text{mA}$	I_C (mA) at $I_E = 6\text{mA}$	I_C (mA) at $I_E = 10\text{mA}$
1	0.0			
2	0.5			
3	1.0			
4	1.5			
5	2.0			
6	4.0			
7	6.0			
8	8.0			
9	10.0			
10	15.0			
11	20.0			

6. Part E & F: Common Collector (CC) Configuration

6.1 Theoretical Background

In the Common Collector (CC) configuration (Emitter Follower), the collector terminal is common to both input (base-collector) and output (emitter-collector) ports.

- **Input Characteristics:** Plot of Base Current (I_B) vs Base-Collector Voltage (V_{BC}) at constant V_{EC} . As V_{BC} increases (increasing reverse bias), I_B decreases because $V_{BE} = V_{EC} - V_{BC}$ decreases for a fixed V_{EC} .
- **Output Characteristics:** Plot of Emitter Current (I_E) vs V_{EC} at constant I_B . Since $I_E = (\beta + 1)I_B \approx I_C$, the curves closely mirror the CE output characteristics.
- **Key Parameters:**

$$\text{Input Resistance: } r_i = \left. \frac{\Delta V_{BC}}{\Delta I_B} \right|_{V_{EC}=\text{const}} \quad (\text{Very High: } \approx 100 \text{ k}\Omega - 500 \text{ k}\Omega) \quad (7)$$

$$\text{Output Resistance: } r_o = \left. \frac{\Delta V_{EC}}{\Delta I_E} \right|_{I_B=\text{const}} \quad (\text{Very Low: } \approx 20\text{--}100 \text{ }\Omega) \quad (8)$$

$$\text{Current Gain: } \gamma = \left. \frac{\Delta I_E}{\Delta I_B} \right|_{V_{EC}=\text{const}} \quad (= \beta + 1 \approx 50\text{--}300) \quad (9)$$

6.2 Circuit Schematic and Simulation Waveforms

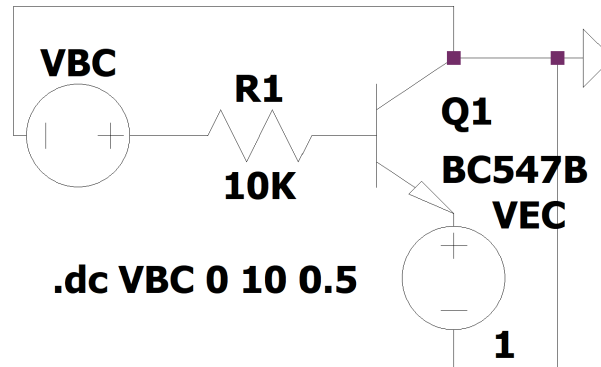
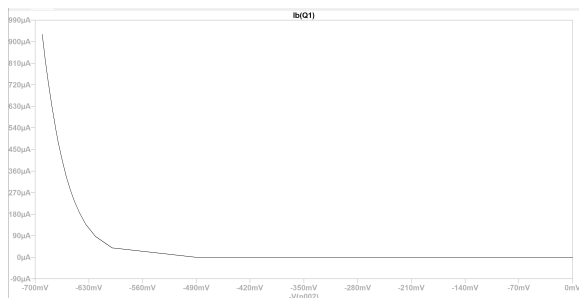
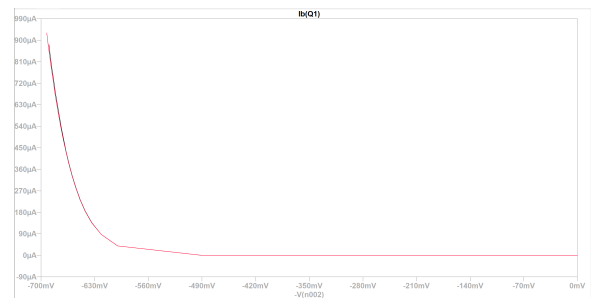


Figure 7: LTspice Circuit Schematic for Common Collector (CC) Configuration.

6.2.1 CC Input Characteristics Simulation



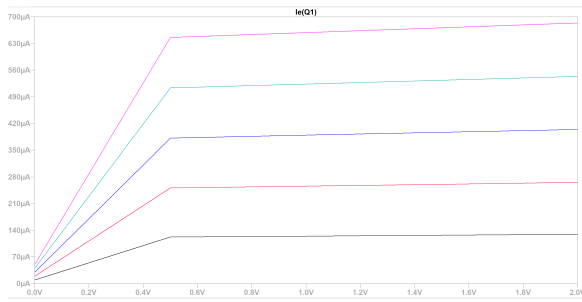
(a) I_B vs V_{BC} at fixed $V_{EC} = 1 \text{ V}$



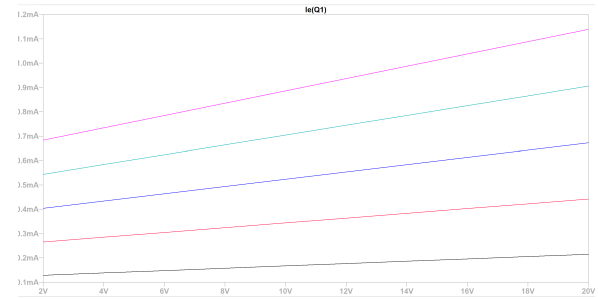
(b) I_B vs V_{BC} at fixed $V_{EC} = 5 \text{ V}, 10 \text{ V}$

Figure 8: LTspice Simulation of CC Input Characteristics (I_B vs V_{BC}).

6.2.2 CC Output Characteristics Simulation



(a) Saturation / Low Voltage Region ($V_{EC} = 0-2$ V)



(b) Active Region ($V_{EC} = 2-20$ V)

Figure 9: LTspice Simulation of CC Output Characteristics (I_E vs V_{EC}) for $I_B = 20 \mu\text{A}, 60 \mu\text{A}, 100 \mu\text{A}$.

6.3 Experimental Observation Tables (Hardware Data)

Table 6: CC Input Characteristics Observation Table (I_B vs V_{BC})

S.No.	V_{BC} (V)	I_B (μA) at $V_{EC} = 1\text{V}$	I_B (μA) at $V_{EC} = 5\text{V}$	I_B (μA) at $V_{EC} = 10\text{V}$
1	0.0			
2	1.0			
3	2.0			
4	3.0			
5	4.0			
6	5.0			
7	6.0			
8	7.0			
9	8.0			
10	9.0			
11	10.0			

Table 7: CC Output Characteristics Observation Table (I_E vs V_{EC})

S.No.	V_{EC} (V)	I_E (mA) at $I_B = 20 \mu\text{A}$	I_E (mA) at $I_B = 60 \mu\text{A}$	I_E (mA) at $I_B = 100 \mu\text{A}$
1	0.0			
2	0.5			
3	1.0			
4	1.5			
5	2.0			
6	4.0			
7	6.0			
8	8.0			
9	10.0			
10	15.0			
11	20.0			

7. Calculations and Verification of Relations

7.1 Formulae for Parameter Extraction

1. Common Emitter Parameters:

$$\beta = \frac{\Delta I_C}{\Delta I_B} \quad (\text{at const } V_{CE}), \quad r_{i(\text{CE})} = \frac{\Delta V_{BE}}{\Delta I_B} \quad (\text{at const } V_{CE}), \quad r_{o(\text{CE})} = \frac{\Delta V_{CE}}{\Delta I_C} \quad (\text{at const } I_B) \quad (10)$$

2. Common Base Parameters:

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \quad (\text{at const } V_{CB}), \quad r_{i(\text{CB})} = \frac{\Delta V_{EB}}{\Delta I_E} \quad (\text{at const } V_{CB}), \quad r_{o(\text{CB})} = \frac{\Delta V_{CB}}{\Delta I_C} \quad (\text{at const } I_E) \quad (11)$$

3. Common Collector Parameters:

$$\gamma = \frac{\Delta I_E}{\Delta I_B} \quad (\text{at const } V_{EC}), \quad r_{i(\text{CC})} = \frac{\Delta V_{BC}}{\Delta I_B} \quad (\text{at const } V_{EC}), \quad r_{o(\text{CC})} = \frac{\Delta V_{EC}}{\Delta I_E} \quad (\text{at const } I_B) \quad (12)$$

7.2 Inter-relationship Verification

The theoretical relations between the current gains are given by:

$$\beta_{calc} = \frac{\alpha}{1 - \alpha} \quad \text{and} \quad \gamma_{calc} = \beta + 1 \quad (13)$$

Table 8: Comparison of Calculated vs Directly Measured Parameters

Parameter / Gain	Directly Measured	Calculated from Formula	% Error
Common Base Gain (α)		—	—
Common Emitter Gain (β)		$\beta_{calc} = \frac{\alpha}{1 - \alpha} =$	
Common Collector Gain (γ)		$\gamma_{calc} = \beta + 1 =$	

7.3 Comparative Summary of BJT Configurations

Table 9: Comparison of Key Parameters across Configurations

Parameter	Common Emitter (CE)	Common Base (CB)	Common Collector (CC)
Input Resistance (r_i)	Medium ($\approx 1 \text{ k}\Omega$)	Low ($\approx 20 \text{ }\Omega$)	High ($\approx 30 \text{ k}\Omega$)
Output Resistance (r_o)	Medium ($\approx 50 \text{ k}\Omega$)	Very High ($\approx 1 \text{ M}\Omega$)	Low ($\approx 5 \text{ }\Omega$)
Current Gain	High ($\beta \approx 100$)	Low ($\alpha < 1$)	High ($\gamma \approx 100$)
Voltage Gain	High	High	≈ 1 (Unity)
Phase Shift	180°	0°	0°
Primary Application	Audio / General Amplification	High Frequency Amplification	Impedance Matching

8. Results and Conclusion

1. The input and output characteristics of the BC547 NPN transistor were simulated in LTspice and studied for Common Emitter (CE), Common Base (CB), and Common Collector (CC) configurations.
2. The schematics and simulation waveforms for each case were integrated and analyzed.

3. The observation tables are prepared for logging experimental hardware readings. Upon completing physical lab testing, calculated parameters $(r_i, r_o, \alpha, \beta, \gamma)$ will be evaluated to verify the theoretical relationships $\beta = \frac{\alpha}{1-\alpha}$ and $\gamma = \beta + 1$.